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Does Change in the Information Environment Affect Financing Choices?

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Abstract. Using brokerage mergers and closures as natural experiments, we examine how exogenous changes in the information environment affect a firm's financing choice. Our difference-in-differences approach shows that exogenous increases in information asymmetry lead firms to substitute away from equity and public debt toward bank debt. Firms with higher risk tend to substitute equity for bank debt, and firms with lower risk tend to substitute bonds for bank debt. The effect of the change in the information environment on a firm's financing choice is more pronounced for firms with worse information environments, such as those with few initial analysts and younger firms. We demonstrate that the mechanism of the change is through a reduction of the issuance of equity and bonds but with an increase of the issuance of bank loans. Further analysis reveals that such firms tend to reduce long-term borrowing, reduce their issuance of subordinated debt, and increase their revolving credit lines.

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Keywords: information environment • analyst coverage • financing choices

1. Introduction

What drives firms' financing choices is an important topic in finance. In this paper, we focus on information asymmetry and examine the effect of an exogenous change in the firm-specific information environment on a firm's financing choice, using two natural experiments and a large sample of firms. Our focus on the impact of information asymmetry on these financing choices is motivated by a large body of theoretical research in which information asymmetry is the primary driving force behind a firm's optimal choice of capital structure (e.g., Leland and Pyle 1977, Campbell and Kracaw 1980, Diamond 1984, Myers and Majluf 1984, Fama 1985, Berlin and Loeys 1988, Diamond 1991, Rajan 1992, Bolton and Freixas 2000, Park 2000). The theoretical arguments generally suggest that firms should rely more on financing choices that are less information sensitive when information asymmetry is high.

The three main financing choices are bank borrowing, bonds, and equity. The theoretical literature suggests that banks are less sensitive than public debt holders and equity holders to information asymmetry, as they have a greater ability to access private information (Fama 1985, James and Smith 2000), stronger

incentives to monitor borrowers (Diamond 1984, Berlin and Loeys 1988, Diamond 1991, Houston and James 1996), and higher flexibility in renegotiating private debt contracts (Gertner and Scharfstein 1991, Park 2000, Denis and Mihov 2003). Banks are more efficient and effective at monitoring and dealing with bankruptcy costs and less sensitive to information changes than equity and public debt holders (Leland and Pyle 1977, Diamond 1984, Boyd and Prescott 1986, Houston and James 1996, Denis and Mihov 2003). When information asymmetry increases (decreases), firms should substitute away from more (less) public information-sensitive financing instruments (equity and bonds) and toward less (more) information-sensitive instruments (bank debts).

Despite the extensive theoretical literature, further empirical evidence of the effect of information asymmetry on financing choice is required.¹ Most studies focus on the determinants of cross-sectional differences in debt structure across firms, and use small, manually collected samples. As with most cross-sectional studies, potential endogeneity problems such as omitted variables or reverse causality can complicate or bias any interpretation of the findings.² Moreover, the proxy variables used in the literature may not perfectly capture

information asymmetry.³ These factors may explain previous mixed findings. Houston and James (1996) find a statistically significant negative relation between growth options and the reliance on bank debt for firms with a single bank relationship. As information asymmetry and agency problems tend to be more severe for firms with greater growth opportunities, “this finding is inconsistent with the hypothesis that firms with larger information asymmetries or greater agency problems of debt rely more on bank financing” (Houston and James 1996, pp. 1864–1865). Using accounting quality as a measure of information asymmetry, Bharath et al. (2008) find that firms with lower accounting quality have a higher proportion of private debt. However, using another information asymmetry measure (favorable private information), they find no robust pattern. In a study of 297 firms, Krishnaswami et al. (1999) use residual stock return volatility as a proxy for information asymmetry and find that those with higher residual volatility have a higher proportion of private debt. Denis and Mihov (2003) find that firms with more fixed assets (indicating a lower level of information asymmetry) rely more on public debt financing. In a recent study, Lin (2016) shows that an increase in collateral value causes firms to rely more on bank loans. This evidence shows that tangible assets capture both information asymmetry and collateral value, which may lead to erroneous conclusions. Hadlock and James (2002) find a positive link between stock return volatility and the likelihood of announcing a new bank loan contract, whereas they document a negative relation between market-to-book ratio and the likelihood of a new bank loan announcement (relative to new public debt issuance).

In this paper, we attempt to address these issues and make the following contributions to the literature. First, we directly investigate the effect of changes in information asymmetry (caused by exogenous changes in analyst coverage) on the changes in financing choice among equity, bond, and bank financing. To alleviate the endogeneity concern that analysts may choose to cover higher-quality firms (Chung and Jo 1996), our identification strategy focuses on two natural experiments pioneered in previous studies—brokerage closures (Kelly and Ljungqvist 2012) and brokerage mergers (Hong and Kacperczyk 2010)—that generate exogenous changes in analyst coverage by reducing the number of analysts following specific firms. These experiments directly affect the analyst coverage of firms but are exogenous to their individual choices of either public or private debt. Kelly and Ljungqvist (2012) show that brokerage closures are driven by business strategy, not by the characteristics of the firms they cover. Wu and Zang (2009) and Hong and Kacperczyk (2010) suggest two reasons for the exogenous drop in analyst coverage resulting from brokerage mergers. First, analysts in the target firms are usually fired for efficiency reasons when more than one analyst is

covering the same firm. Second, analysts may leave if they are unwilling to face the uncertainty of integration and adapt to a culture change. News of brokerage closures and mergers can quickly reach firms and investors through press releases and media outlets. A key advantage of the selected identification strategy is that it not only resolves concerns about endogeneity, but also alleviates the problem of omitted variables by allowing multiple shocks to affect different firms at different times.

Second, we construct our panel data set of corporate debt structure by taking advantage of recently available databases such as Standard & Poor’s (S&P) Capital IQ, which contains extensive debt structure data for more than 60,000 firms worldwide. Our extensive sample allows for a comprehensive, large-scale study. The rich capital structure data allow us to examine the various types of debt chosen by firms, such as credit lines, term loans, senior bonds, and subordinate bonds, and a variety of debt maturities. We also examine the information channels and find that the effect is stronger when firms face a poor information environment.

Third, we directly test our key underlying assumption that banks are less sensitive than public debt holders to changes in information asymmetry. To the best of our knowledge, we are the first to examine the information sensitivity of public and private debt instruments. From portfolio matching and difference-in-differences (DID) estimates, we gauge the information sensitivity of banks by examining the changes in bank loan spreads before and after an exogenous reduction in analyst coverage. Bank loan data are compiled from DealScan, a Thomson Reuters Loan Pricing Corporation database containing loan information for U.S. and foreign corporations. Using the recently available daily bond transaction data in the Trade Reporting and Compliance Engine (TRACE) data set, we use an event study approach to examine the instantaneous bond market reaction (measured by the cumulative abnormal returns) to the exogenous reduction in analyst coverage, and thus gauge the information sensitivity of bondholders to the change in information asymmetry. Our empirical findings strongly confirm the hypothesis that bondholders are much more sensitive to public information than banks. The results provide direct support for our main finding of a link between a change in the information environment and a change in corporate debt structure.

Following previous studies, we successfully identify 23 brokerage closures and mergers between 2003 and 2010. These exogenous events are associated with 795 treated firm-year observations. The control sample consists of firms from the same industry (two-digit Standard Industrial Classification (SIC) code) and year, of the same *size* and *Q*, and from the same analyst coverage (*Anacov*) tercile as the treatment firm (i.e., the firm that experienced an exogenous decrease in analyst coverage). To ensure that we are capturing only the

effect due to the exogenous shock to analyst coverage, we compare the change in financing choice of the treatment firm from one year prior to the brokerage exit ($t - 1$) to one year after ($t + 1$) with the changes in the choices of the control firms, while controlling for a wide range of other factors.⁴ We find that bank loan percentages increase significantly for firms that experience a drop in analyst coverage, whereas their public bond percentages and equity portions decrease significantly. The evidence suggests that on average, an increase in information asymmetry causes firms to switch from public borrowing and equity to private borrowing. After our treated firms experience an exogenous drop in analyst coverage, their bank loan percentages increase (decrease) 0.797 percentage points more (less) than those of the control firms, their public bond percentages decrease (increase) 0.613 percentage points more (less), and their equity portions decrease (increase) by 1.326 percentage points more (less). Given that the sample mean bank loan percentage is 7.516%, the sample mean public bond percentage is 14.417%, and the sample mean equity portion is 41.934%, our results are economically significant, indicating a bank loan increase of 10.604% (0.797/7.516) of the mean, a public bond decrease of 4.252% (0.613/14.417) of the mean, and an equity decrease of 3.162% (1.326/41.934) of the mean.

We demonstrate the robustness of our main results through various approaches including a DID matching approach (average treatment effect on the treated (ATT)) and an extended sample. We also provide evidence that treated firms borrow more in bank loans but issue fewer bonds and less equity. The issuance evidence is consistent with the findings of Purnanandam and Rajan (2018),⁵ who study the impact of growth option exercise on capital structure and show that leverage is negatively correlated with their proxy for growth option conversion. They use brokerage house mergers as exogenous shocks to firms' information environment to further demonstrate that the negative relationship becomes stronger when the information environment of a firm deteriorates. Their Table 10 provides evidence that firms affected by a brokerage house merger issue less equity and more debt than control firms after the shock. Purnanandam and Rajan (2018) focus mainly on the impact of growth option exercise and do not investigate bank loans and bonds separately, whereas we focus on the direct effects of information asymmetry changes on the structures of bank loans, public bonds, and equity. We provide evidence that bank loans and bonds are affected differently by the shocks: treated firms tend to reduce their long-term borrowing and issuance of subordinated debt, and increase their revolving credit lines.

Our results thus establish a causal relationship between a reduction in analyst coverage and a change in financing choice. The effect should be stronger when the change in analyst coverage is more likely to cause

increased information asymmetry, aggravate agency costs, and increase the risk of moral hazard. If the change in financing choice is due to the change in analyst coverage, we expect a more profound effect for firms with lower initial analyst coverage, younger firms, and those with high idiosyncratic risk, as these firms experience a more dramatic change in their information environment. Our cross-sectional tests confirm these predictions. We also find that for firms with low cash flow risk, the change in financing choice is marked by a reduction in bonds and an increase in bank loans, whereas for firms with high cash flow risk, the change is marked by a reduction in equity and an increase in bank loans. This evidence is broadly consistent with Bolton and Freixas's (2000) framework, which suggests that riskier firms prefer bank loans, less risky firms use bonds, and those in between use both equity and bonds. Less risky firms therefore choose to reduce bonds and higher-risk firms choose to reduce equity when information asymmetry increases.

Next, we consider changes in firms' choice of debt type at a more molecular level and find that the decrease in bonds is mainly in subordinated bonds, and the increase in bank loans is mainly in revolving credit lines. In terms of debt maturity, we find that long-term borrowing decreases significantly. We also examine the different levels of information sensitivity of public versus private debt holders, which is the key assumption underlying our empirical analysis. Using daily bond returns compiled from the TRACE data set, we document a significant negative abnormal return on bonds for firms that experience a drop in analyst coverage. Over the $[-10, +10]$ trading window, the cumulative abnormal bond return is around -10 basis points, indicating an increase in the required rate of public bonds traded in the secondary market after an increase in information asymmetry. We further divide our sample into subsamples based on low and high analyst coverage before the natural experiments, and observe negative bond cumulative abnormal returns (CARs; -20 basis points) for firms with low initial analyst coverage. CARs for firms in the high-analyst-coverage subsample are, however, not significantly different from zero. We then address the potential change in bank financing and examine bank loan spreads before and after the exogenous reduction in analyst coverage, using DID regression analysis.⁶ In contrast to the control firms, we find no significant change in bank loan spread for the treated firms. Taken together, the empirical analyses are consistent with our underlying assumption and explain why the treated firms may switch among the available financing choices.

2. Literature Review and Hypothesis Development

Extensive theoretical research suggests that information asymmetry is the primary driving force in a firm's

optimal choice of capital structure (e.g., Leland and Pyle 1977, Campbell and Kracaw 1980, Diamond 1984, Myers and Majluf 1984, Fama 1985, Berlin and Loeys 1988, Diamond 1991, Rajan 1992, Bolton and Freixas 2000, Park 2000). Equity financing bears no bankruptcy costs but carries higher dilution costs for firms with higher future cash flows, because they are selling claims on future cash flows. Firms weigh the bankruptcy costs and dilution costs in an information asymmetry environment when they consider equity financing. Those with a lower risk of bankruptcy and that can display this lower risk to the market are more likely to choose equity financing than debt financing. When information asymmetry increases, the dilution costs of firms also increase because they are less likely to reveal their risk levels. The desirability of equity financing will thus decrease.

Unlike equity financing, debt financing in this framework bears few dilution costs but is associated with bankruptcy costs. However, bond financing involves relatively inefficient liquidation costs compared with bank debt financing, whereas the cost of bank loans is higher because of the additional intermediation costs (banks need to raise capital themselves). In the framework provided by Bolton and Freixas (2000) and in other theoretical studies (e.g., Houston and James 1996, Park 2000, Hadlock and James 2002), debt holders of different types differ in their seniority in cash flow claims, monitoring incentive, information advantage, and ability to price opaque assets. The literature suggests that banks are less sensitive to information asymmetry than public debt holders. First, banks have better access to the private information and future prospects of borrowers than public debt investors (Fama 1985, James and Smith 2000). Hadlock and James (2002) point out that banks have a comparative advantage in pricing the securities of firms with greater information opacity. Second, banks have more concentrated ownership of debt claims and thus have stronger incentives than public debt holders to monitor borrowers. In contrast, the dispersed ownership of public bondholders aggravates free-rider problems and reduces the individual bondholder's incentive to engage in costly monitoring (Diamond 1984, Berlin and Loeys 1988, Diamond 1991, Houston and James 1996).⁷ Third, it is easier to renegotiate or restructure private debt contracts than publicly traded contracts.⁸ Banks have greater flexibility in contract renegotiation, so they can discipline firms more effectively and efficiently than bondholders (Gertner and Scharfstein 1991, Park 2000, Denis and Mihov 2003). The combination of superior access to private information, stronger monitoring incentives, and greater flexibility in renegotiation makes banks more efficient and effective at monitoring and dealing with bankruptcy costs and less sensitive to information changes than public debt holders (Leland and Pyle 1977, Diamond 1984, Boyd and Prescott 1986, Houston and James 1996,

Denis and Mihov 2003). Banks are therefore less sensitive to changes in public information because they are significantly less reliant on the information from publicly available sources.⁹ Based on the above arguments, we propose the following hypothesis.

Hypothesis 1. *Firms that experience a loss of analysts should rely more on bank borrowings and less on bond financing and equity financing after the loss of analysts, compared to the control firms.*

Empirical evidence for the effect of information asymmetry on financing choice is limited in the theoretical literature. The research into debt choice is also mixed (see the detailed discussion of the empirical findings in the introduction). To empirically test the hypothesis concerning the effects of information asymmetry on firms' financing choices more effectively, we must identify the exogenous shocks affecting firms' information environment. Hong and Kacperczyk (2010) and Kelly and Ljungqvist (2012) introduce such natural experiments and show that brokerage house mergers and closures bring exogenous shocks to this information environment. Financial analysts play an important role in reducing the information asymmetry between firms and external investors, by distributing both public and private information to institutional investors and millions of individual investors through research reports and media outlets such as newspapers and TV programs (Miller 2006, Ellul and Panayides 2018).¹⁰ An exogenous decrease in analyst coverage thus provides a clear measure of change in the corporate information environment.¹¹ By adopting such a setting, we can provide evidence to test our hypothesis.

3. Sample, Data, and Variables

To measure changes in the information environment, we rely on two natural experiments that are widely used in the literature and that create exogenous shocks to analyst coverage. Financial analysts provide in-depth analyses of financial statements and other corporate disclosures.¹² In addition to the mandatory disclosures, they allocate real resources and effort to activities such as interviewing managers, consumers, and suppliers, and visiting companies and facilities to produce informative research reports (Brown et al. 2015). By distributing both public and private information to institutional and individual investors through various channels, analysts effectively mitigate the information asymmetry between firms and external investors.¹³ Compared with other proxies for information asymmetry, analyst coverage is less controversial and is widely used and accepted in the literature (e.g., Givoly and Lakonishok 1979, Lys and Sohn 1990, Lang and Lundholm 1996, Gu and Wu 2003, Frankel and Li 2004, Chen and Lin 2017, Kecskes et al. 2017, Merkley et al. 2017). An exogenous decrease in analyst coverage thus

provides a clean measure of change in the corporate information environment—that is, an exogenous increase in information asymmetry. Table IA1 in the online appendix shows a decrease in liquidity (i.e., an increase in the Amihud (2002) measure) and an increase in bid–ask spread, thus confirming that the exogenous analyst coverage loss does indeed lead to an increase in information asymmetry.¹⁴

To capture this exogenous change in the information environment, we construct our sample by identifying firms that lost financial analysts because of brokerage closures and mergers, which are exogenous to the corporate policies of the individual firms. The first source of exogenous change—brokerage closures—occurred because the revenue from traditionally profitable operations of brokerages, including trading, market making, and investment banking, has diminished since the 2000s. The research operations of brokerages produced common goods and were very costly, so they terminated their research operations in response to the reduced profits from other operations. Brokerages thus ended their research operations for strategic reasons and not because of the firms they cover, so brokerage closures are ideal, exogenous events, uncorrelated with firms' choices of debt structure. Kelly and Ljungqvist (2012) use brokerage closures as natural experiments on the supply of information to test the role of information asymmetry on asset pricing.

The second source of exogenous change in the information environment is brokerage mergers. When two brokers merge, they often fire analysts to avoid the duplication of roles and culture clashes (Wu and Zang 2009). If both brokerage houses have an analyst following the same firm, the analyst from the target firm is usually dismissed (Hong and Kacperczyk 2010). Therefore, firms followed by both brokerage houses before the merger will experience an exogenous drop in analyst coverage. By taking brokerage mergers as natural experiments, Hong and Kacperczyk (2010) find concrete evidence of the information environment change, showing an increase in analysts' forecast bias due to the exogenous reduction in competition among analysts.

These two types of natural experiments allow us to resolve potential endogeneity concerns and also alleviate omitted variable bias by allowing multiple shocks to affect different firms at different times. Using the same natural experiments, Chen et al. (2015) find causal effects of financial analysts on mitigating agency issues. As our debt structure data are from S&P Capital IQ, we restrict the sample to events occurring after 2002, when the coverage became comprehensive.¹⁵ The brokerage closures and mergers between 2003 and 2008 are from Kelly and Ljungqvist (2012). We extend the events list to 2010, and search for brokers who disappeared from the Institutional Brokers' Estimate System (I/B/E/S) database between 2008 and 2010. We then use news published by

Factiva to confirm that these disappearances were due to brokerage closures. We then extend the brokerage mergers list following Hong and Kacperczyk (2010). We use Thomson's SDC Mergers and Acquisition database to identify acquisitions between brokerage houses. Specifically, we restrict mergers to the 2008–2010 period and select only the acquirers and targets with primary SIC codes of 6211 or 6282. We retain only completed deals in which 100% of the target was acquired. We then match the brokerage house with the I/B/E/S data to obtain the brokerage code and compare the active period of the brokerage house to ensure that the code is correct.

After extending the list of experiments, we construct a sample of firms that experienced an exogenous drop in analyst coverage, using different strategies for brokerage closures and brokerage mergers, following the literature. For brokerage closures, we identify a sample of covered firms using the I/B/E/S unadjusted historical data set. Following Kelly and Ljungqvist (2012), we retain only firms for which the estimate in I/B/E/S "stops" after the date of the brokerage disappearance. For brokerage mergers, we retain only those firms covered by both brokerage houses before the merger but only one analyst after the merger. The latter requirement rules out the possibility that the brokerage dropped both analysts due to firm-specific characteristics. To ensure that there are no other confounding events and the event is not a temporary analyst loss, we require a realized reduction in analyst coverage of the treated firm during the six months after the natural experiment. After merging with our debt structure data, our final event list includes 10 brokerage closures and 13 brokerage mergers from 2003 to 2010. We use Factiva to identify the announcement dates of the events, when investors could infer that analyst coverage had been terminated exogenously.

To better capture the financing choices of the firm, we combine data from Compustat and S&P Capital IQ data. The total assets and common equity data are obtained from Compustat. The detailed debt structure data are from S&P Capital IQ. Bank loan amount is defined as the sum of revolving credit and term loans. Public bond amount is defined as the sum of senior and subordinated bonds and notes.¹⁶ We also compare the difference between total debt from Capital IQ and that from Compustat and find the data difference to be very small. Bank loan percentage is defined as the ratio of bank loans to total assets. Public bond percentage is defined as the ratio of public bonds to total assets. Equity percentage is defined as the ratio of common equity to total assets. To purge the data and exclude potential data error, we exclude all observations for which the sum of bank loan percentage, bond financing, and equity financing is greater than 100%. We define year +1 as the first fiscal year after the shock and year –1 as the most recent fiscal year in which the year-end date is before the event month. Firms without events are

extracted as potential control firms for the regression analysis and matching sample analysis.

Analyst coverage data are drawn from I/B/E/S, and analyst coverage is defined as the number of unique analysts following a firm each year. Financial and accounting data are taken from Compustat. Potential control firms for regression analysis and matching sample analysis are first required to be in the same industry (two-digit SIC code) as the treated firms. They must then match the *size*, *Q*, and analyst coverage terciles of the treated firms within each industry and each year.¹⁷

For our main regression analysis, each treated firm is matched with at most five control firms.¹⁸ Where there are more than five matches based on the above screening procedure, we retain the five with the greatest similarity to the treated firm by calculating the differences in *size*, *Q*, and analyst coverage of the treated and control firms. We then assign the rank of the absolute difference for each of these three variables and sum the three ranks to obtain the total rank. The five (at most) control firms with the lowest total rank are used for the main regression analysis. Following the literature, we include *size*, *Q*, return on assets (ROA), *tangibility*, and *rating* as control variables. The variable definitions are presented in Table A.1 in the appendix.

Table 1 presents the summary statistics for our baseline regression model. Our sample consists of 6,330 firm-year observations from 1,683 unique U.S. public firms over the 2002–2012 period. We have 1,590 treated firm-year observations (for each of the 795 treated firm-year, there are two year observations to capture before and after the

event) from 651 unique U.S. public firms (a firm may experience more than one instance of analyst reduction in the sample period). For our dependent variables, on average, equity accounts for 41.934% of total assets, bank loans for 7.516% of total assets, and public bonds for 14.417% of total assets. The dependent variables exhibit considerable variations. The standard deviations of bank loan, public bond, and equity percentages to total assets are 11.323%, 15.599% and 25.628%, respectively. The large variations in these dependent variables highlight the importance of understanding information channels' potential impact on firms' financing choices.

4. Main Analysis: The Effect of a Change in the Information Environment on a Firm's Financing Choice

4.1. Balance Test

To demonstrate the validity of our matching process, we conduct a balance test to show that the treated firms and matched control firms are similar in their firm fundamentals in year $t - 1$. Specifically, we test whether variables such as *Bank/AT%*, *Bond/AT%*, *Equity/AT%*, *Anacov*, *Size*, *Q*, *ROA*, *Tangibility*, and *rating* for treated firms and control firms are statistically equal at mean. The results presented in Table 2 show that the differences in firm fundamentals are generally not statistically significant, confirming the validity of our matching process.

4.2. Regression Results

We plot the financing choices for treated and matched control firms around the event time—that is, from

Table 1. Summary Statistics

Variable	No. of observations	Mean	Standard deviation	25th percentile	Median	75th percentile
<i>Bank/AT%</i>	6,330	7.516	11.323	0.000	1.741	10.265
<i>Bond/AT%</i>	6,330	14.417	15.599	0.197	10.131	23.178
<i>Equity/AT%</i>	6,330	41.934	25.628	19.776	40.745	61.476
<i>Size</i>	6,330	8.244	1.744	6.895	8.187	9.523
<i>Q</i>	6,330	1.823	0.841	1.146	1.552	2.205
<i>Anacov</i>	6,330	13.173	7.093	7.000	12.000	18.000
<i>ROA</i>	6,330	0.116	0.081	0.051	0.112	0.172
<i>Tangibility</i>	6,330	0.410	0.370	0.087	0.306	0.689
<i>Rating</i>	6,330	5.043	1.844	3.000	5.000	7.000
Δ Bank loan/lagged assets%	6,330	0.828	4.048	−0.249	0.000	1.410
<i>Bond issuance amount/lagged assets%</i>	6,330	0.396	1.213	0.000	0.000	0.000
<i>Equity issuance amount/lagged assets%</i>	6,330	0.757	5.590	0.000	0.000	0.000

Notes. This table reports descriptive statistics for our financing choice, firm characteristics, and control variables for the sample used in the regression analysis. Our sample covers 6,330 unique firm-years over the 2002–2012 period, from 1,683 unique U.S. public firms. *Bank/AT%* is defined as the amount of revolving credit and term loans as a percentage of total assets. *Bond/AT%* is defined as the amount of senior and junior bonds as a percentage of total assets. *Equity/AT%* is defined as the amount of common equity as a percentage of total assets. *Size* is defined as log of total assets of a firm. A firm's *Q* is defined as market value of assets divided by book value of total assets. *Anacov* is the number of analysts following the firm this year. *ROA* is calculated as operating income before depreciation divided by total assets. *Tangibility* is calculated as the gross value total property, plant, and equipment divided by total assets. *Rating* is the S&P credit rating, ranging from 1 to 7. Δ Bank loan/lagged assets% is the change in total bank loans scaled by lagged total assets. *Bond issuance amount/lagged assets%* is the amount of new bond issuance scaled by lagged total assets. *Equity issuance amount/lagged assets%* is the amount of new equity issuance scaled by lagged total assets.

Table 2. Balance Test

	Mean		Difference	<i>p</i> -Value	Kolmogorov–Smirnov two-sample test <i>p</i> -value
	Treated	Control			
<i>Bank/AT%</i>	6.746	6.441	0.305	0.452	0.804
<i>Bond/AT%</i>	15.345	14.864	0.481	0.428	0.639
<i>Equity/AT%</i>	43.401	43.336	0.064	0.949	0.813
<i>Size</i>	8.621	8.539	0.082	0.221	0.204
<i>Q</i>	1.831	1.800	0.031	0.320	0.589
<i>Anacov</i>	15.069	14.998	0.071	0.799	0.160
<i>ROA</i>	0.117	0.119	−0.003	0.362	0.002***
<i>Tangibility</i>	0.406	0.416	−0.010	0.479	0.190
<i>Rating</i>	4.804	4.808	−0.004	0.952	0.807

Notes. This table reports the balance test results. The sample consists of 4,184 firm-years from 1,683 unique publicly traded U.S. firms covering the 2002–2012 period. The 795 firm-years that experience analyst coverage loss in year t are assigned as treated. We locate at most five control firms for each treated firm according to *size*, Q , and analyst coverage; 3,389 firm-years in the sample are assigned as the control sample. We keep duplicated control firm-year observations if such firm-year observations are matched to different treated firm-year observations. *Bank/AT%* is defined as the amount of revolving credit and term loans as a percentage of total assets. *Bond/AT%* is defined as the amount of senior and junior bonds as a percentage of total assets. *Equity/AT%* is defined as the amount of common equity as a percentage of total assets. *Size* is defined as log of total assets of a firm. A firm's Q is defined as market value of assets divided by book value of total assets. *Anacov* is the number of analysts following the firm this year. *ROA* is calculated as operating income before depreciation divided by total assets. *Tangibility* is calculated as the gross value total property, plant, and equipment divided by total assets. *Rating* is the S&P credit rating, ranging from 1 to 7.

***Statistically significant at the 1% level.

year $t - 3$ to year $t + 3$. Figure 1 gives the responses of the treated firms' equity component and debt components to the exogenous loss of analyst following. We find a sharp decrease in equity and public bond percentage and a slight increase in bank loan percentage from year t to year $t + 2$. The figure also confirms the parallel trend in debt structure for treated and control firms before the event—that is, from year $t - 3$ to year t .

We then run a pooled DID regression to empirically test and quantify the effect of an exogenous drop in analyst coverage on a firm's financing choice. Our regression specification controls for characteristics that have been shown to affect a firm's financing choice (Houston and James 1996, Denis and Mihov 2003, Lin et al. 2013).

$$y_{i,t} = \alpha + \beta_1(Treated_{i,t} \times Post_{i,t}) + \beta_2 Treated_{i,t} + \beta_3 Post_{i,t} + \delta' X_{i,t-1} + \varepsilon_{i,t}, \quad (1)$$

where $y_{i,t}$ is bank loan percentage or public bond percentage, $Treated_{i,t}$ is a dummy variable equal to one if the firm has experienced an exogenous drop in analyst coverage due to brokerage closures or mergers and zero otherwise, and $Post_{i,t}$ is a dummy variable equal to one in year +1 and zero in year −1. Our variable of interest is the interaction variable, $Treated_{i,t} \times Post_{i,t}$. The coefficient β_1 , which captures the DID effect, shows the difference in debt choice between the treated and control firms after the natural experiments. The vector X_i contains a set of firm-specific variables, including *size*,

Tobin's Q, *ROA*, *tangibility*, and *rating*. Details of the variables are presented in Table A.1 in the appendix. We also control for firm fixed effect and year fixed effect in the regression analysis.

Table 3 shows our main results using DID regressions. Columns (1)–(3) report the ordinary least squares regression results for the effects of exogenously reduced analyst coverage on bank loan percentage, public bond percentage, and equity percentage, respectively.

The results in columns (1)–(3) are consistent with our hypothesis. As shown in column (1), the bank loan percentages of treated firms increase (decrease), on average, 0.797 percentage points more (less) than those of their matched control firms. Given that the sample mean bank loan percentage is 7.516%, the economic significance is relatively large (0.797/7.516 is 10.604%). The public bond percentages of treated firms decrease (increase) 0.613 percentage points more (less) than those of their matched control firms. The percentage point difference is 4.252% of the mean (14.417%). Finally, their equity portions decrease (increase) 1.326 percentage points more (less) than those of their matched control firms, representing a percentage point difference of 3.162% of the mean (41.934%). The finding of reduction in equity financing experienced by treated firms after the shocks is consistent with the finding in Derrien and Kecskes (2013). The coefficients for the interaction term, $Treated_i \times Post_{i,t}$, are significant at the 10% level or better for all our main specifications. From the results, we can confirm that when faced with increased information

Figure 1. (Color online) Financing Choices of Treated and Control Firms Around the Event Year



Notes. The figure shows the changes in financing choice for our treated and control firms around the event year. Bank Loan/Assets is defined as the amount of revolving credit and term loans as a percentage of total assets, Bond/Assets is defined as the amount of senior and junior bonds as a percentage of total assets, and Equity/Assets is defined as the amount of common equity as a percentage of total assets.

asymmetry, firms shy away from equity and bond financing and switch to bank loan financing, which is less sensitive to information asymmetry issues.

4.3. Robustness

Various robustness checks, including a DID matching estimator approach (ATT) and an extended sample regression, confirm our main finding that a change in the information environment leads to a change in the financing choice.

Table 3. Impact of an Exogenous Drop in Analyst Coverage on Firms' Financing Choices

	(1) Bank/AT%	(2) Bond/AT%	(3) Equity/AT%
<i>Treated</i> × <i>Post</i>	0.797*** [0.292]	−0.613* [0.319]	−1.326*** [0.442]
<i>Treated</i>	−0.417* [0.241]	0.678* [0.377]	0.628 [0.520]
<i>Post</i>	0.247 [0.411]	0.743* [0.410]	−1.796 [1.081]
<i>Size</i>	−1.525* [0.765]	−5.025*** [1.112]	−13.175*** [1.815]
<i>Q</i>	−0.171 [0.305]	0.184 [0.471]	3.098*** [0.933]
<i>ROA</i>	1.592 [3.514]	−1.980 [4.832]	65.793*** [10.224]
<i>Tangibility</i>	6.471*** [2.477]	4.616 [3.200]	1.390 [6.103]
<i>Rating</i>	0.549 [0.384]	−0.610 [0.566]	−0.291 [0.576]
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations	6,330	6,330	6,330
Adjusted <i>R</i> ²	0.758	0.817	0.853

Notes. This table reports the results of the DID regressions examining the effect of an exogenous drop in financial analysts on firms' financing choices. Our sample consists of 6,330 firm-years from 1,683 publicly traded U.S. firms covering the 2002 to 2012 period. Our sample includes 1,590 treated firm-years. Control firms are matched according to industry, year, size, and *Q* tercile. We take the closest five control firms if the treated firms are matched with more than five control firms. Bank/AT% is defined as the amount of revolving credit and term loans as a percentage of total assets. Bond/AT% is defined as the amount of senior and junior bonds as a percentage of total assets. Equity/AT% is defined as the amount of common equity as a percentage of total assets. *Treated* is a dummy variable equal to 1 if the firm has experienced an exogenous drop in analyst coverage and 0 otherwise. *Post* is a dummy equal to 1 for the year after the shock and 0 otherwise. Definitions of other variables are given in the appendix. The estimations correct the error structure for heteroskedasticity and within-firm and year error clustering, with standard errors reported in brackets.

*Statistically significant at the 10% level; ***statistically significant at the 1% level.

4.3.1. DID Matching Estimator (ATT). We locate the nearest nontreated neighbor for each of our treated firms based on our selection criteria, as described in Abadie and Imbens (2006). Specifically, we require our treated and control firms to be matched by industry (two-digit SIC code) and year. We further require them to be of the same size, *Q*, and analyst coverage tercile.

Having obtained a sample for matching, we then attempt to find a set of control firms for each of our treated firms according to different matching dimensions. We use various combinations of matching dimensions, drawn from size, *Q*, ROA, tangibility (property, plant, and equipment), rating, analyst coverage, and pre-event capital structure. For each of our treated firms, we find five nearest-neighbor control firms and compare the difference in the changes in bank loan percentage and

public bond percentage from year $t - 1$ to year $t + 1$ between the treated and control firms.

Table 4 presents the matching results for bank loan percentage, public bond percentage, and equity percentage across different matching dimensions. Panel A shows that across all our matching dimensions, there is a significant increase in bank loan percentage among the treated firms compared with their matched control firms. For example, by matching the treated firms with control firms according to *size*, *Q*, and analyst coverage, we obtain an ATT of 0.890, which is significant at the 1% level. The difference in bank loan percentage for our treated firms is 0.874 percentage points, compared with 0.008 for the matched control firms. The other matching dimensions yield similar results, confirming that treated firms increase their bank loan percentage after experiencing an exogenous drop in analyst coverage. Panel B

in Table 4 shows the matching results for public bond percentage. In the sample matched by *size*, *Q*, and analyst coverage, we find a significant ATT of -0.546 with a *t*-statistic of -2.24 . Our treated firms show a reduction in public bonds to total assets of -0.496 percentage points, whereas there is an increase for control firms of 0.017 percentage points. All other matching dimensions provide supportive evidence with little variation in magnitude and statistical significance. In panel C, we present the response of treated firms in equity financing. Consistent with our regression results, we find a reduction of 1.580 percentage points of equity financing in treated firms, after matching with similar control firms. The ATT matching results are all significant at the 1% level in panel C. The nearest-neighbor matching results lend additional support to our aforementioned empirical results.

Table 4. Impact of an Exogenous Drop in Analyst Coverage on Firms' Financing Choices—Evidence from Nearest-Neighbor Matching

Panel A: Bank/AT%					
Matching variables	No. of observations	Bank_Diff treated	Bank_Diff control	ATT (treated vs. control)	<i>t</i> -stat.
<i>Size/Q/Anacov</i>	795	0.874	0.008	0.890***	4.66
<i>Size/Q/Anacov/ROA/Tangibility</i>	795	0.872	0.048	0.846***	4.47
<i>Size/Q/Anacov/ROA/Tangibility/Rating</i>	795	0.872	0.124	0.793***	4.12
<i>Size/Q/Anacov/ROA/Tangibility/Rating/PreBank/PreBond/PreEquity</i>	795	0.874	0.226	0.921***	5.04
Panel B: Bond/AT%					
	No. of observations	Bond_Diff treated	Bond_Diff control	ATT (treated vs. control)	<i>t</i> -stat.
<i>Size/Q/Anacov</i>	795	-0.496	0.017	-0.546^{**}	-2.24
<i>Size/Q/Anacov/ROA/Tangibility</i>	795	-0.497	0.114	-0.645^{**}	-2.73
<i>Size/Q/Anacov/ROA/Tangibility/Rating</i>	795	-0.497	0.097	-0.650^{**}	-2.73
<i>Size/Q/Anacov/ROA/Tangibility/Rating/PreBank/PreBond/PreEquity</i>	795	-0.496	0.254	-0.562^{**}	-2.52
Panel C: Equity/AT%					
	No. of observations	Equity_Diff treated	Equity_Diff control	ATT (treated vs. control)	<i>t</i> -stat.
<i>Size/Q/Anacov</i>	795	-4.794	-3.225	-1.580^{***}	-3.87
<i>Size/Q/Anacov/ROA/Tangibility</i>	795	-4.798	-2.958	-1.816^{***}	-4.46
<i>Size/Q/Anacov/ROA/Tangibility/Rating</i>	795	-4.798	-3.073	-1.720^{***}	-4.24
<i>Size/Q/Anacov/ROA/Tangibility/Rating/PreBank/PreBond/PreEquity</i>	795	-4.802	-2.882	-1.385^{***}	-3.53

Notes. This table presents the DID estimates (ATT) for the changes in bank loan percentage, public bond percentage, and equity percentage following the exogenous reductions in analyst following. *Bank/AT%* is defined as the amount of revolving credit and term loans as a percentage of total assets. *Bond/AT%* is defined as the amount of senior and junior bonds as a percentage of total assets. *Equity/AT%* is defined as the amount of common equity as a percentage of total assets. ATT is the Abadie and Imbens (2006) bias-corrected average treated effect matching estimator. Panels A, B, and C report the matching estimates for change in bank loan percentage, change in public bond percentage, and change in equity percentage, respectively. Bank_Diff, Bond_Diff, and Equity_Diff treated and control are the differences in *Bank/AT%*, *Bond/AT%*, and *Equity/AT%*, respectively, before and after the natural experiment for treated and control firms. *t*-stat. refers to *t*-statistics for the ATT estimate. We require the treated firms and matched control firms to be in the same industry, same year, and of the same *size*, *Q*, analyst coverage tercile. For each treated firm, we find the five nearest control firms according to different matching criteria. *PreBank*, *PreBond*, and *PreEquity* are the *Bank/AT%*, *Bond/AT%*, and *Equity/AT%*, respectively, in the year before the exogenous reduction in analyst following. Other variable definitions are given in the appendix. Heteroskedasticity-consistent *t*-statistics are reported.

*Statistically significant at the 10% level; ***statistically significant at the 1% level.

4.3.2. Extended Sample Period and Removing Sample Firms After 2008. Our main regression results (Table 3) contain only two years of observations for each firm: year $t - 1$ and year $t + 1$. However, the financing choices of the firms are likely to be sticky. Therefore, in this subsection, we test the impact of exogenously reduced analyst coverage on firms' financing choices using an extended sample—that is, from year -3 to year $+3$. Extending the sample allows us to demonstrate the long-term effect of the natural experiment and also to test whether there is any pre-event trend, which serves as a placebo test and can further validate our natural experiment. The regression results for the extended sample are reported in Table IA2 of the online appendix.

In the extended sample regression, year -1 is the base year and is omitted in the regression. Years -3 , -2 , $+1$, $+2$, and $+3$ are year dummies for the time before and after the event. To save space, we report only the interaction terms in the table. We find that in the years after the exogenous loss of financial analysts, there is a lasting change in firms' financing choices. The interactions between *Treated* and *Year*₊₁, *Treated* and *Year*₊₂, and *Treated* and *Year*₊₃ are significant in all of our regressions, confirming the effect of an exogenous information environment change on firms' financing choices over a longer period. More importantly, we find no significant coefficients for the interactions between *Treated* and the year dummies before the event. This placebo test verifies the validity of our using exogenous analyst coverage loss as a natural experiment.

In this paper, we have extended the list of brokerage mergers and closures to 2008, 2009, and 2010. Some of these mergers and closures were probably due to the financial crisis, which suggests that firms that lost analysts also lost underwriters and other important relationships. We thus exclude this period to ensure it does not affect our findings, and conduct an analysis without sample firms after 2008. The findings reported in Table IA3 of the online appendix demonstrate that our main findings are robust and are not driven by this particular period.¹⁹

4.3.3. Evidence from Issuance of Bank Loans, Bonds, and Equity. To further establish the link between exogenous analyst coverage loss and firms' choice between public debt and bank loans, we investigate the mechanism of the change in financing choice by examining the treated firms' issuance of bank loans, bonds, and equity before and after the events. Our findings are reported in Table 5. Columns (1)–(3) report the findings on the issuance of bank loans, bonds, and equity, respectively, based on the construction of continuous variables of issuance data. We would ideally investigate the bank loan originations using DealScan, but it is difficult to obtain

a clean measure of bank loan originations because it contains only large syndicated loans. We instead use the change in total bank loan amount (from Capital IQ) adjusted by lagged year total assets (the change in bank debt amount from $t - 1$ to t scaled by assets in year $t - 1$) as a proxy variable. Our findings in column (1) show that the exogenous loss of analyst coverage leads to an increase in the bank loan amount. For bond and equity issuance, we retrieve the data from SDC Platinum. We add the bond and equity issuance amounts for each firm for each year and further adjust the total bond and total equity issuance amounts by the firms' total assets in the lagged year. The regression results are reported in columns (2) and (3) of Table 5. The exogenous loss of analyst coverage leads to a decrease in the new bond issuance amount. Similarly, we find a reduction in equity issuance after the exogenous loss of financial analysts. As shown in Table 1, the issuance variables have some zero values. To address the concern of nonlinearity, we report the findings based on a dummy variable design for the issuance data. For bank loans, we define the loan issuance dummy as one if there is an increase in total bank loans and zero otherwise. For corporate bonds and equity, the issuance dummy is defined as one if there is an issuance and zero otherwise. Columns (4)–(6) indicate that the findings using dummy variables are consistent with those in column (1)–(3). By demonstrating an increase in bank loan issuance and a decrease in bond and equity issuance, we confirm that the increased information asymmetry leads firms to shift away from public bonds and equity toward bank loans.

5. Further Exploration

The above evidence indicates that after an exogenous drop in analyst coverage, firms tend to use more bank loans and fewer equity and public bonds, which suggests that a change in the information environment causes a change in a firm's choice of financing. We next investigate the mechanisms by which information asymmetry affects financing choice.

5.1. Effect of Information Channel

Intuitively, if our main findings are driven by information asymmetry, the effect should be stronger when the asymmetry is more likely to be adversely affected. The subsample analysis is conducted by partitioning our full sample by the initial number of analysts following, *firm age*, and *firm idiosyncratic risk*.²⁰ First, the role of a single analyst is more important when a firm is followed by fewer analysts. According to Hong and Kacperczyk (2010), after an exogenous drop in analyst coverage, the analyst forecast bias is more substantial for firms with low initial analyst coverage than those with high initial coverage. Consequently, the increase in information asymmetry

Table 5. Impact of an Exogenous Drop in Analyst Coverage on Issuance

	Continuous			Issuance dummy		
	Bank	Bond	Equity	Bank	Bond	Equity
<i>Treated</i> × <i>Post</i>	0.368** [0.171]	−0.108* [0.059]	−0.620** [0.284]	0.093*** [0.021]	−0.027* [0.015]	−0.024* [0.012]
<i>Treated</i>	−0.122 [0.145]	0.028 [0.068]	0.397 [0.285]	−0.048** [0.022]	0.005 [0.015]	0.019 [0.011]
<i>Post</i>	0.259 [0.254]	0.051 [0.042]	0.173 [0.233]	0.004 [0.030]	0.006 [0.014]	0.017* [0.009]
<i>Size</i>	−1.572*** [0.493]	−0.090 [0.096]	−2.726*** [0.790]	−0.046 [0.037]	0.013 [0.028]	−0.072*** [0.021]
<i>Q</i>	0.411* [0.226]	0.064 [0.064]	0.753* [0.372]	0.007 [0.027]	0.016 [0.014]	0.012 [0.009]
<i>ROA</i>	6.226*** [2.134]	3.078*** [0.833]	−6.644** [2.963]	0.190 [0.246]	0.623*** [0.188]	−0.477*** [0.136]
<i>Tangibility</i>	0.240 [1.773]	0.955** [0.383]	2.772 [3.669]	0.045 [0.128]	0.189** [0.086]	0.128 [0.079]
<i>Rating</i>	0.187 [0.177]	−0.038 [0.054]	0.050 [0.106]	−0.016 [0.018]	−0.013 [0.016]	0.011* [0.006]
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,330	6,330	6,330	6,330	6,330	6,330
Adjusted <i>R</i> ²	0.093	0.213	0.219	0.259	0.268	0.240

Notes. This table reports the effect of an exogenous drop in analyst coverage on new bank loan, bond, and equity issuance. Bank loan issuance data are retrieved from the Capital IQ database. Bond issuance and equity issuance amounts are obtained from the SDC database. In cases where there are several issuances during a year, we calculate the total amount. For continuous variable design, bank loan issuance is defined as the change in total bank loan amount (from Capital IQ) adjusted by lagged year total assets (the change in bank debt amount from $t - 1$ to t scaled by lagged assets). We adjust the total amount of bond and equity issuance by the firms' lagged total assets. Regarding the dummy variable design, for bank loans, we define a loan issuance dummy to be 1 if there is an increase in total bank loans and 0 otherwise. For corporate bond and equity, the issuance dummy is defined as 1 if there is an issuance and 0 otherwise. *Treated* is a dummy variable equal to 1 if the firm has experienced an exogenous drop in analyst coverage and 0 otherwise. *Post* is a dummy equal to 1 for the year after the shock and 0 otherwise. Other variable definitions are given in the appendix. The estimations correct the error structure for heteroskedasticity and within-firm and year error clustering, with standard errors reported in brackets.

*Statistically significant at the 10% level; **statistically significant at the 5% level; ***statistically significant at the 1% level.

should be greater for firms with low initial analyst coverage. To test the effect of the level of initial analyst coverage on our results, we partition our sample by the level of initial analyst coverage and report the regression results in panel A of Table 6. Specifically, we divide the sample into firms followed by fewer than five analysts versus those followed by five or more analysts.

The results show that our main findings are driven by the subsample of firms with low initial analyst coverage, as the interaction coefficients of our regressions on the bond, equity, and bank loan percentages are all statistically significant for the subsample with low initial analyst coverage.²¹ For the subsample of firms with high initial analyst coverage, the changes in bank, bond, and equity percentages are not statistically significant. We also test the differences in the coefficients on *Treated* × *Post* across the different subsamples. The tests indicate that there are statistical differences between the low- and

high-analyst-following subsamples for bank and equity financing choices.

Second, *firm age* can be related to the information environment. We define *firm age* as the number of years since the firm's initial public offering (IPO). Our findings based on the partition of firms by age are presented in panel B of Table 6. The findings are driven by the subsample of younger firms, which exhibit significant changes in terms of financing choices. No statistically significant changes in financing choices are found in the subsample of older firms. We also test the differences in the coefficients on *Treated* × *Post* across the different subsamples. All three tests indicate statistically significant differences in the financing choices between the young and old subsamples.

Third, *firm idiosyncratic risk* is defined as the standard deviation of the residual from the Fama-French three-factor model, following Ang et al. (2006). This

Table 6. Impact of an Exogenous Drop in Analyst Coverage on Firms' Financing Choices—The Role of the Information Channel

	Panel A: Initial analyst coverage by five analysts					
	<i>Bank/AT%</i>		<i>Bond/AT%</i>		<i>Equity/AT%</i>	
	<5	≥5	<5	≥5	<5	≥5
<i>Treated</i> × <i>Post</i>	2.457* [1.335]	0.377 [0.235]	−1.249* [0.699]	−0.227 [0.361]	−4.384** [1.913]	−0.779 [0.473]
<i>Treated</i> × <i>Post</i> difference						
Difference	2.080**		−1.022		−3.605**	
<i>p</i> -value	0.05		0.31		0.05	
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,112	7,256	1,112	7,256	1,112	7,256
Adjusted <i>R</i> ²	0.720	0.789	0.879	0.837	0.825	0.861
	Panel B: Firm age					
	<i>Bank/AT%</i>		<i>Bond/AT%</i>		<i>Equity/AT%</i>	
	Young	Old	Young	Old	Young	Old
<i>Treated</i> × <i>Post</i>	1.536*** [0.426]	−0.146 [0.258]	−1.390* [0.840]	0.388 [0.462]	−1.810** [0.773]	−0.472 [0.691]
<i>Treated</i> × <i>Post</i> difference						
Difference	1.682***		−1.778***		−1.337**	
<i>p</i> -value	0.00		0.00		0.02	
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,832	4,524	3,832	4,524	3,832	4,524
Adjusted <i>R</i> ²	0.768	0.784	0.843	0.857	0.844	0.879

Notes. This table reports the results of DID regressions, examining the effect of an exogenous drop in financial analysts on firms' financing choices, conditional on the number of analyst following before the shock and firm age. In panel A, the sample is partitioned by five initial analysts following, and in panel B, the sample is partitioned by firm age. *Bank/AT%* is defined as the amount of revolving credit and term loans as a percentage of total assets. *Bond/AT%* is defined as the amount of senior and junior bonds as a percentage of total assets. *Equity/AT%* is defined as the amount of common equity as a percentage of total assets. *Treated* is a dummy variable equal to 1 if the firm has experienced an exogenous drop in analyst coverage and 0 otherwise. *Post* is a dummy variable equal to 1 for the year after the shock and 0 otherwise. Definitions for other variables are given in the appendix. Coefficients on control variables are not reported to conserve space. The estimations correct the error structure for heteroskedasticity and within-firm and year error clustering, with standard errors reported in brackets.

*Statistically significant at the 10% level; **statistically significant at the 5% level; ***statistically significant at the 1% level.

variable also captures a firm's information channel. This variable has been used previously to capture information asymmetry (Bhagat et al. 1985, Blackwell et al. 1990). When *idiosyncratic risk* is higher, firms face more information asymmetry. Our findings based on the partition of *idiosyncratic risk* are presented in Table IA5 of the online appendix, to conserve space. We find that the pattern in the full sample is driven by the firms with high *idiosyncratic risk*, which exhibit significant changes in all three financing choices. No statistically significant changes in financing choices are found in the low *idiosyncratic risk* subsample. We also test the differences in the coefficients on *Treated* × *Post* across the different subsamples. All three tests

indicate that there are statistical differences between the two subsamples.

We also partition the sample by tangible assets, and the untabulated results indicate some limited evidence that the documented pattern in the full sample is concentrated in the subsample of low *tangibility* firms. Specifically, we find that the decreases in bond and equity financing are statistically significant in the subsample of low *tangibility* firms. However, increases in bank loan financing are observed in both subsamples. While firms with more fixed assets can indicate a lower level of information asymmetry, Lin (2016) shows that an increase in tangible assets (collateral value) causes firms to rely more on bank loans. *Tangibility* not only captures the information

channel effect, but also captures firms' accessibility to bank loan financing in our research setting, which can explain our findings here.

5.2. Firms with Different Risk Levels

In Bolton and Freixas's (2000) model, firms' optimal capital structure under an information asymmetry environment is related to their risk level. We extend their model by partitioning our sample into firms with different risk levels. Bolton and Freixas (2000) predict that the safest firms use bonds, riskier firms use bank debt, and those in between use both bonds and equity. Following this equilibrium, we expect firms with low risk (corresponding to the safest firms in Bolton and Freixas's (2000) model) to reduce the percentage of bonds and increase the percentage of bank debt in their capital structure, and firms with high risk (corresponding to the middle firms in the model) to reduce the percentage of equity and increase the percentage of bank debt in their capital structure.²² We proxy firms' cash flow risk using earnings volatility and *cash flow volatility*, following Lee et al. (2017). As shown in Table 7, we find that for the firms with low risk, the change in the information environment generally reduces their bond financing and increases their bank loan financing. These firms tend to have a large percentage of bond financing due to their low risk before the information environment change. However, for firms with high risk, we find that the change mainly reduces equity financing and increases bank loan financing. Intuitively, such firms may not have much bond financing before the information environment change. This pattern is generally consistent with Bolton and Freixas's (2000) theoretical prediction. We also test the differences in the coefficients on *Treated* × *Post* across the low- and high-risk subsamples in the table. Four out of six comparisons show a statistically significant difference of 10% or better.

5.3. Exploring Debt Structure

The previously mixed findings on the effect of information asymmetry on public bonds versus private bank loans and our rich debt structure data lead us to further explore the impact of increased information asymmetry on debt structure and debt issuance.

S&P Capital IQ classifies debt into commercial paper, revolving credit, term loans, senior bonds and notes, subordinated bonds and notes, capital leases, and other debt, enabling researchers to provide additional evidence of debt structure. Colla et al. (2013) use this data set to provide one of the first large-sample studies of the patterns and determinants of debt structure. We also perform an analysis of specific debt structure components to further reveal the impact of changes in the information environment. In Table 8, we present

the results for various components of the debt structure, to understand the mechanism of such a debt structure change in more detail.²³ As revealed in columns (1) and (2) of the table, the increase in bank loans is driven by revolving credit lines, while the change in *term loans* is not statistically significant. This finding is consistent with literature highlighting the importance of credit lines (e.g., Ivashina and Sharfstein 2010, Cornett et al. 2011, Colla et al. 2013, Acharya et al. 2014).²⁴

The results reported in columns (3) and (4) show that the decrease in bonds is driven by *subordinated bonds*, whereas the change in *senior bonds* is not statistically significant. This further confirms the effect of information asymmetry on debt structure in our main results. Senior bonds are less risky than subordinated bonds, and an increase in information asymmetry has a greater effect on riskier bonds. Holders of riskier types of bonds are more concerned about the increased uncertainty from greater information asymmetry than those of more senior bonds.

Our analysis of the role of debt maturities in column (5) indicates that treated firms reduce their use of long-term borrowing after the information environment change. Firms tend to rely more on debt maturing in five years or less when the information environment is negatively affected. Research suggests that debt maturity plays an important role in the debt structure. Almeida et al. (2011) find that debt maturity can significantly affect investment decisions. They find that firms with a large portion of *long-term debt* set to mature immediately after the third quarter of 2007 cut their investment-to-capital ratio by 2.5 percentage points more (on a quarterly basis) than otherwise similar firms whose debt was scheduled to mature after 2008. Choi et al. (2017) find that newly issued bond maturities complement preexisting bond maturities and conclude that firms spread out maturity dates over time to reduce rollover risks. Our findings on the effect of increased information asymmetry on firms' debt maturities in this setting are consistent with the notion that treated firms regarded the events as short-term shocks and reacted by adopting more short-term, rather than long-term, borrowing. Our findings on reduced *long-term debt* by treated firms after the shocks are also consistent with findings in Derrien and Kecskes (2013).

In addition, we explore the effect of a change in information asymmetry on another important source of corporate financing, trade credit. Firms can obtain financing from trade credit. We use three variables to proxy for trade credit, that is, (1) total accounts payable scaled by total liabilities, (2) total accounts payable scaled by total assets, and (3) total accounts receivable scaled by total assets. The empirical results are presented in Table 9. We do not find any significant impact of the change in information environment

Table 7. Impact of an Exogenous Drop in Analyst Coverage on Firms' Financing Choices—The Role of Cash Flow Risk

	Bank/AT%		Bond/AT%		Equity/AT%	
	Low	High	Low	High	Low	High
Panel A: EPS volatility						
<i>Treated</i> × <i>Post</i>	0.846*** [0.248]	0.751* [0.407]	−1.286** [0.548]	0.407 [0.383]	0.559 [0.468]	−3.827*** [1.096]
<i>Treated</i>	−0.574 [0.355]	−0.001 [0.353]	0.512 [0.596]	0.442 [0.444]	−0.285 [0.802]	1.807** [0.733]
<i>Post</i>	0.255 [0.334]	−0.411 [0.341]	0.425 [0.559]	−0.561 [0.401]	−3.062** [1.075]	−4.408** [1.675]
<i>Size</i>	1.560 [1.729]	1.928 [1.390]	−3.788 [2.958]	−0.026 [2.148]	−2.790 [4.000]	1.891 [2.066]
<i>Q</i>	−0.406 [0.668]	−0.265 [0.449]	−1.543* [0.871]	0.717 [0.959]	3.223** [1.345]	3.852* [1.983]
<i>ROA</i>	−18.324* [8.912]	−7.798 [5.397]	−36.819** [14.746]	−1.913 [7.777]	30.042* [15.366]	74.952*** [17.887]
<i>Tangibility</i>	7.525 [5.252]	1.300 [3.685]	−10.830* [6.041]	0.418 [6.151]	−33.792*** [8.634]	−8.245 [7.654]
<i>Rating</i>	0.329 [0.411]	0.676 [0.608]	−1.308 [0.780]	1.535* [0.748]	−0.073 [0.587]	0.529 [0.711]
<i>Treated</i> × <i>Post</i> difference						
Difference						
<i>p</i> -value		0.095 0.35		−1.693*** 0.00		4.386*** 0.00
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,292	4,018	4,292	4,018	4,292	4,018
Adjusted <i>R</i> ²	0.772	0.793	0.828	0.866	0.877	0.843
Panel B: Cash flow volatility						
<i>Treated</i> × <i>Post</i>	0.852*** [0.228]	0.750* [0.411]	−0.810** [0.325]	−0.056 [0.712]	0.564 [0.928]	−3.602** [1.608]
<i>Treated</i>	−0.474 [0.333]	−0.212 [0.334]	0.116 [0.483]	0.596 [0.607]	0.113 [0.872]	1.782 [1.346]
<i>Post</i>	0.000 [0.433]	−0.207 [0.247]	0.080 [0.313]	−0.258 [0.635]	−2.470** [1.024]	−5.047** [1.725]
<i>Size</i>	1.665 [1.043]	2.435* [1.341]	0.057 [1.508]	−1.134 [2.826]	−3.669 [2.869]	2.255 [2.523]
<i>Q</i>	−0.966** [0.423]	−0.469 [0.510]	0.937 [0.930]	−1.172 [0.873]	4.523*** [1.312]	2.404 [1.494]
<i>ROA</i>	1.209 [14.787]	−15.900*** [5.217]	−22.407 [15.811]	−9.613 [10.196]	21.605 [28.855]	67.119*** [12.490]
<i>Tangibility</i>	11.198 [6.494]	2.533 [4.396]	0.978 [6.708]	−2.734 [5.518]	−19.836** [8.521]	−17.737* [8.773]
<i>Rating</i>	−0.044 [0.406]	0.816 [0.570]	−0.222 [0.286]	−0.285 [1.272]	−1.484** [0.547]	1.598* [0.791]
<i>Treated</i> × <i>Post</i> difference						
Difference						
<i>p</i> -value		0.102 0.36		−0.754* 0.08		4.166*** 0.00
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,218	4,092	4,218	4,092	4,218	4,092
Adjusted <i>R</i> ²	0.819	0.737	0.895	0.796	0.921	0.795

Notes. This table reports the results of the DID regressions, examining the effect of an exogenous drop in financial analysts on firms' financing choice, conditional on the cash flow risk level before the shock. We split the sample into two subsamples according to whether the treated firms are classified as high or low cash flow risk (earnings per share (EPS) volatility and cash flow volatility). *Bank/AT%* is defined as the amount of revolving credit and term loans as a percentage of total assets. *Bond/AT%* is defined as the amount of senior and junior bonds as a percentage of total assets. *Equity/AT%* is defined as the amount of common equity as a percentage of total assets. *Treated* is a dummy variable equal to 1 if the firm has experienced an exogenous drop in analyst coverage and 0 otherwise. *Post* is a dummy variable equal to 1 for the year after the shock and 0 otherwise. Definitions for other variables are given in the appendix. The estimations correct the error structure for heteroskedasticity and within-firm and year error clustering, with standard errors reported in brackets.

*Statistically significant at the 10% level; **statistically significant at the 5% level; ***statistically significant at the 1% level.

Table 8. Impact of an Exogenous Drop in Analyst Coverage on Firms' Debt Structure

	(1) Revolving credit	(2) Term loans	(3) Subordinated bonds	(4) Senior bonds	(5) Long-term debt
<i>Treated</i> × <i>Post</i>	0.453** [0.158]	0.043 [0.207]	−0.329* [0.190]	0.226 [0.345]	−1.383*** [0.385]
<i>Treated</i>	−0.251 [0.219]	0.046 [0.168]	0.046 [0.121]	0.371 [0.366]	0.927** [0.403]
<i>Post</i>	−0.037 [0.214]	0.338 [0.213]	−0.011 [0.070]	0.611* [0.321]	−1.165** [0.429]
<i>Size</i>	−0.801 [0.534]	−0.269 [0.388]	−0.771*** [0.237]	−2.916** [1.164]	3.567** [1.604]
<i>Q</i>	−0.029 [0.182]	−0.201 [0.257]	−0.101 [0.131]	0.079 [0.448]	−0.893* [0.423]
<i>ROA</i>	−0.837 [2.817]	2.453 [2.171]	−2.001 [1.228]	0.938 [4.992]	10.452 [6.106]
<i>Tangibility</i>	3.366* [1.929]	1.216 [1.509]	−0.748 [0.959]	2.220 [2.615]	−15.932*** [5.210]
<i>Rating</i>	0.320 [0.282]	0.071 [0.132]	−0.142 [0.111]	−0.278 [0.525]	0.402 [0.694]
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	6,330	6,330	6,330	6,330	4,133
Adjusted R^2	0.732	0.664	0.812	0.813	0.675

Notes. This table reports results of the DID regressions examining the effect of an exogenous drop in financial analysts on firms' debt components. Specifically, we examine the effect of an exogenous drop in financial analysts on the portions of revolving credit, term loans, subordinated bonds, and senior bonds, scaled by total assets. We define *long-term debt* as the amount of debt maturing in more than five years, scaled by total assets. *Treated* is a dummy variable equal to 1 if the firm has experienced an exogenous drop in analyst coverage and 0 otherwise. *Post* is a dummy equal to 1 for one year after the shock and 0 for the year before the shock. Definitions of other variables are in the appendix. The estimations correct the error structure for heteroskedasticity and within-firm and year error clustering, with standard errors reported in brackets.

*Statistically significant at the 10% level; **statistically significant at the 5% level; ***statistically significant at the 1% level.

on treated firms' trade credit financing after the events. One possible explanation is that suppliers have superior information about firms, and they are less sensitive to the change in firms' information environment.²⁵

5.4. Evidence from Bond Returns and Bank Loan Pricing

Our DID regression and matching results show that after experiencing an exogenous drop in analyst coverage, the debt structure of the treated firms exhibits a higher bank loan percentage and lower public loan percentage than that of the control firms. These findings are consistent with our main argument that sensitivity to the information environment differs across debt holders, with bondholders being most sensitive to exogenous changes in the information environment among debt holders. In this section, we provide more direct evidence by examining bond market reactions and bank loan pricing. Lacking access to private information and expertise in pricing opaque assets, bondholders have more concerns regarding potential moral hazard issues brought about by an exogenous increase

in information asymmetry. They therefore perceive the loss of analyst coverage due to a brokerage exit negatively, and the cost of the firm's public bonds should increase. We consequently expect a negative cumulative return from the bond market. Banks are, however, less affected by potential moral hazard problems and less sensitive to changes in the information environment. We thus predict no significant change in bank loan pricing.

To examine the bond market reaction for the treated firms, we use the daily bond trading data from the recently available TRACE database, which provides bond identification information including date, time of execution, price, yield, and trading volume. We construct the daily price of the bond following Bessembinder et al. (2009). In calculating the trade-volume-weighted price, we require the trade volume to be greater than \$100,000. We then follow Easton et al. (2009) to calculate the Treasury-adjusted bond returns for each bond, focusing on abnormal bond returns over the trading window of $[-10, +10]$, as the bond market is rather illiquid. We calculate

Table 9. Impact of an Exogenous Drop in Analyst Coverage on Trade Credit

	(1) AP/TL	(2) AP/AT	(3) AR/AT
<i>Treated</i> × <i>Post</i>	−0.080 [0.272]	−0.119 [0.161]	0.189 [0.147]
<i>Treated</i>	0.021 [0.242]	0.139 [0.139]	−0.033 [0.120]
<i>Post</i>	−0.545 [0.327]	−0.090 [0.163]	−0.219 [0.201]
<i>Size</i>	−1.297* [0.717]	−0.384 [0.351]	−1.088*** [0.301]
<i>Q</i>	0.438* [0.249]	0.123 [0.125]	0.137 [0.268]
<i>ROA</i>	7.526** [2.767]	−1.327 [1.713]	1.120 [2.891]
<i>Tangibility</i>	−2.041 [1.694]	0.565 [0.792]	−0.380 [1.070]
<i>Rating</i>	0.232 [0.225]	0.138 [0.134]	0.247 [0.253]
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations	6,236	6,244	6,297
Adjusted <i>R</i> ²	0.968	0.986	0.973

Notes. This table presents the impact of an exogenous drop in analyst coverage on trade credit. Trade credit is proxied by total accounts payable (AP; Compustat Item 70) scaled by total liabilities (TL) in column (1), by total accounts payable (Compustat Item 70) scaled by total assets (AT) in column (2), and by total accounts receivable (AR; Compustat Item 2) scaled by total assets in column (3). *Treated* is a dummy variable equal to 1 if the firm has experienced an exogenous drop in analyst coverage and 0 otherwise. *Post* is a dummy equal to 1 for the year after the shock and 0 otherwise. Other variable definitions are in the appendix. The estimations correct the error structure for heteroskedasticity and within-firm and year error clustering, with standard errors reported in brackets.

*Statistically significant at the 10% level; **statistically significant at the 5% level; ***statistically significant at the 1% level.

the buy-and-hold abnormal bond return with the following equation:

$$BR_{i,-m,+n} = \frac{BP_{i,+n} - BP_{i,-m}}{BP_{i,-m}}, \quad (2)$$

where $BR_{i,-m,+n}$ is the buy-and-hold bond return, $BP_{i,-m}$ is the first available bond price for bond i in trading window $[-10, -1]$, and $BP_{i,+n}$ is the last bond price for bond i in trading window $[0, +10]$. For firms with multiple bonds trading over this period, we use the bond with the highest liquidity over the previous 12 months. Each bond must have trades in both pre-event window and postevent window to be in the sample. This method identifies 427 bonds with at least two trades over the $[-10, +10]$ trading window. We then adjust the buy-and-hold bond return using the contemporaneous U.S. Treasury returns with the following equation:

$$CAR_{i,-m,+n} = BR_{i,-m,+n} - \prod_{t=-m}^{+n} TR_t, \quad (3)$$

where $CAR_{i,-m,+n}$ is the abnormal bond return over the trading window $[-10, +10]$, and TR_t is the treasury return on the same day. We match the bond with treasuries according to maturity and then choose the one with the closest coupon rate.

Table 10 shows the bond market reaction to firms with an exogenous drop in analyst coverage. As shown in panel A, on average, there is a negative cumulative abnormal bond return over the $[-10, +10]$ trading window with a magnitude of -0.102% , significant at the 10% level. As the actual trading window differs from $[-10, +10]$ because of the illiquidity of bond trading, we also calculate the abnormal bond return per day. The real trading days are defined as $n + m + 1$. Our trading day window varies from 4 to 21, and the abnormal return per day is defined as $CAR_{i,-m,+n}$ divided by $(n + m + 1)$.

Again, we demonstrate the effect of low initial analyst coverage on bond return. We split our sample into two subsamples based on initial analyst coverage. Panel B in Table 10 reports the findings. For the low-analyst-coverage group, the mean bond CAR is -0.199% , and the mean bond CAR per day is -0.011% , both significant at the 5% level. However, for the

Table 10. Impact of an Exogenous Drop in Analyst Coverage on the Cost of Public Bonds—Abnormal Bond Returns

	<i>N</i>	Mean CAR (%)	<i>t</i> -statistic	25th percentile	Median	75th percentile
Variable	Panel A: Cumulative abnormal bond return					
CAR	1,181	−0.102*	−1.92	−0.743	−0.087	0.733
CAR/ <i>days</i>	1,181	−0.004	−1.28	−0.045	−0.005	0.044
	Panel B: Cumulative abnormal bond return partitioned by analyst coverage					
	Low analyst coverage					
CAR	563	−0.199**	−2.41	−0.921	−0.096	0.845
CAR/ <i>days</i>	563	−0.011**	−2.04	−0.055	−0.006	0.054
	High analyst coverage					
CAR	618	−0.013	−0.19	−0.633	−0.066	0.650
CAR/ <i>days</i>	618	0.002	0.40	−0.038	−0.004	0.039

Notes. This table presents the impact of an exogenous drop in analyst coverage on the cost of public bonds. We calculate the cumulative abnormal returns on the bonds of treated firms. Following Bessembinder et al. (2009), we calculate the trading-volume-weighted bond price for each bond each day. We match the bond with treasuries according to maturity and coupon rate. Then we measure the buy-and-hold abnormal return (CAR) for our treated firms' bond over the $[-10, +10]$ trading window. *CAR/days* is CAR divided by the trading days between the first trade in the $[-10, -1]$ trading window and the last trade in the $[0, +10]$ trading window. Panel A reports the results, and panel B reports the bond CAR conditional on the analyst coverage.

*Statistically significant at the 10% level; **statistically significant at the 5% level.

high-analyst-coverage group, the mean bond CAR is -0.013% , insignificant at any conventional level. The evidence directly supports the intuitive assumption that if a firm is followed by a large number of financial analysts, losing one analyst has no significant consequence, as bondholders can obtain enough information from the remaining analysts. However, if a firm is followed by a small number of analysts, the loss of one analyst can lead to a significant decrease in bond return, which increases the cost of public borrowing.²⁶

We also investigate whether an exogenous loss in analyst coverage affects bank loan pricing, using bank loan issuance data from DealScan, which provides detailed information for a large number of commercial loans in the United States.²⁷ We use the all-in-spread drawn as a measure of the interest rate charged on a loan facility. Specifically, we investigate the spread of the largest bank loan issued over the fiscal year in year -1 and year $+1$. We take the natural logarithm of the loan spread to mitigate the effect of skewness (Graham et al. 2008, Lin et al. 2011) and retain only firms with no missing spread data for both year -1 and year $+1$. The results in Table IA6 of the online appendix show no significant coefficients for the interaction terms, indicating no change in bank loan pricing for the treated firms compared with the control firms. These results provide evidence for the key underlying assumption of the choice between public bonds and private bank loans. By showing that there is no change in bank loan pricing but a significant negative bond market reaction, we confirm that banks are less sensitive than bondholders to the exogenous loss of analysts.

We also explore the bank loan syndicate and covenant data. Table IA7 in the online appendix shows the findings and they indicate that treated firms tend to use more lenders and more lead arrangers (statistically significant in the specification controlling for industry fixed effects and year fixed effects), indicating an increased risk sharing after the events. We find no significant changes for the financial covenant number in the loan contracts after the shocks.

6. Conclusion

This paper provides empirical support for the theoretical prediction that the information environment affects a firm's capital structure—the mix of equity, bank debt, and public debt. Specifically, we investigate

the effect of an exogenous increase in information asymmetry on a firm's financing choice, using brokerage mergers and closures as natural experiments. We provide definitive evidence that increased information asymmetry causes firms to substitute away from more information-sensitive debt instruments (public bonds) and equity to those that are less information sensitive (bank loans). In addition, we identify the mechanisms through which a change in the information environment affects a firm's capital choice by providing evidence on the issuance of bonds, bank loans, and equity. Consistent with other studies that use the loss of brokerages as natural experiments, our results are driven by firms with low initial analyst coverage, younger firms, and those with high idiosyncratic risk. We also find that low cash flow risk firms tend to switch from bonds to bank loans, and high cash flow risk firms tend to switch from equity to bank loans. We further study the components of debt instruments and show that the increase in bank loans is primarily due to an increase in credit lines, and the decrease in bonds is primarily due to a decrease in subordinated bonds. We find a significant negative cumulative bond return but no change in bank loan spread after a change in the information environment.

Our paper contributes to the literature on firms' financing choices by providing evidence that firms substitute away from bond financing and equity financing toward bank loan financing and that bond markets react negatively when there is an exogenous reduction in analyst coverage. By illustrating the effect of information asymmetry on a firm's financing choice, our findings provide a potential explanation for the growing reliance on the corporate bond market and the shrinking share of bank loan financing in the credit market over time, which has led to a general improvement in the corporate information environment over the past few decades.

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Appendix.

Table A.1. Variable Definitions

Variable name	Definition
<i>Bank/AT%</i>	Bank loans (sum of revolving credit and term loans) as a percentage of total assets (AT) Source: S&P Capital IQ and Compustat
<i>Bond/AT%</i>	Public bonds (sum of senior bonds and notes and subordinated bonds and notes) as a percentage of total assets Source: S&P Capital IQ and Compustat
<i>Equity/AT%</i>	Common equity as a percentage of total assets $Equity/AT\% = \text{item 60}/\text{item 6}$ Source: Compustat
<i>Size</i>	Log of total assets of a firm $Size = \text{Log}(\text{item 6})$ Source: Compustat
<i>Q: Tobin's Q</i>	Market value of assets divided by book value of total assets $Q = (\text{item 6} - \text{item 60} + \text{item 25} \times \text{item 199})/\text{item 6}$ High market-to-book is a dummy equal to 1 if the treated firm has a <i>Q</i> in the top tercile of our sample; low market-to-book is a dummy equal to 1 if the treated firm has a <i>Q</i> in the bottom tercile of our sample Source: Compustat
<i>ROA</i>	ROA is calculated as operating income before depreciation divided by total assets $ROA = (\text{item 13}/\text{item 6})$ Source: Compustat
<i>Tangibility</i>	<i>Tangibility</i> is calculated as the gross value total property, plant, and equipment divided by total assets $Tangibility = (\text{item 7}/\text{item 6})$ Source: Compustat
<i>Rating</i>	<i>Rating</i> is the S&P credit rating, ranging from 1 to 7. The smaller the number, the lower credit rating; for example, rating 1 represents AAA rating. Speculative rating has a rating above 4 and 7 represents companies without any rating. Source: Compustat
<i>Anacov</i>	Number of analysts following the firm this year Source: I/B/E/S
<i>Firm age</i>	Number of years from the firm's IPO year Source: CRSP
<i>Idiosyncratic risk</i>	The standard deviation of residuals from Fama–French three-factor model (Fama and French 1993). We regress firm daily return on Fama–French three factors every month. For a firm year, we average the monthly idiosyncratic risk. Source: CRSP
<i>EPS volatility</i>	The standard deviation of earnings per share during the past five years Source: Compustat
<i>Cash flow volatility</i>	The standard deviation of cash flow during the past five years Source: Compustat
<i>Bank loan issuance</i>	The change in total bank loans scaled by lagged total assets Source: Capital IQ and Compustat
<i>Bond issuance</i>	The amount of new bond issuance scaled by lagged total assets Source: SDC and Compustat
<i>Equity issuance</i>	The amount of new equity issuance scaled by lagged total assets Source: SDC and Compustat
<i>Revolving credit</i>	The amount of revolving credit scaled by total assets Source: Capital IQ and Compustat
<i>Term loans</i>	The amount of term loans scaled by total assets Source: Capital IQ and Compustat
<i>Subordinated bonds</i>	The amount of subordinated bonds scaled by total assets Source: Capital IQ and Compustat
<i>Senior bonds</i>	The amount of senior bonds scaled by total assets Source: Capital IQ and Compustat
<i>Long-term debt</i>	The amount of debt maturing in over five years, as a percentage of total assets Source: Compustat
<i>Trade credit</i>	<i>Trade credit</i> is proxied by total accounts payable (Compustat item 70) scaled by total liabilities, by total accounts payable (Compustat item 70) scaled by total assets, and by total accounts receivable (Compustat item 2) scaled by total assets, respectively Source: Compustat

Endnotes

¹ A detailed literature review is presented in the next section.

² For instance, firms with public borrowings may have to disclose more information to the public, leading to lower observed information asymmetry.

³ For instance, stock return volatility may proxy for managerial risk-taking activities, market-to-book ratios may proxy for firm performance or the additional value over book assets that debt holders can access in the credit event, and fixed assets may represent a firm's tangibility and debt capacity (Lin 2016).

⁴ We also show that the change in capital structure is beyond one year in the extended sample period study.

⁵ The reduction of equity issuance is also consistent with the results of Chang et al. (2006). They conclude that firms that are subject to greater information asymmetry should issue equity less frequently.

⁶ Conducting an analogous event study using loan trading data would be more informative. However, few academic studies use secondary loan market data. Wittenberg-Moerman (2008), for example, examines how the quality of financial reporting affects secondary loan transactions over a five-year period from 1998 to 2003, whereas our sample period starts from 2003. We therefore use the DealScan loan issuance data and draw on the relatively mature literature on bank loan contracts.

⁷ Even if bondholders were willing to monitor them, the monitoring would be less efficient because of the unnecessary and costly duplication of effort (Houston and James 1996, Lin et al. 2013).

⁸ As required by the Trust Indenture Act of 1939, firms must receive unanimous consent from each bondholder to alter any material term in the bond indenture (Houston and James 1996). See Smith and Warner (1979) for a discussion of this act.

⁹ Although bank debt imposes other costs that might drive borrowers away from bank debt, such as bargaining power over firms' profits due to the developed private information (Sharpe 1990, Rajan 1992) and increased monitoring costs and regulatory costs incurred by banks (Fama 1985, Diamond 1991), when the agency costs of debt are high, bank financing can be less expensive than public debt.

¹⁰ Compared with other market participants, financial analysts have substantial financial and industry knowledge, and, more importantly, they track firms regularly and interface with management directly through earnings announcement conference calls (e.g., Frankel et al. 1999, Chen and Matsumoto 2006, Ke and Yu 2006). Furthermore, financial analysts also commit real resources and effort to activities such as interviewing consumers and suppliers and visiting companies and facilities to gather information beyond the mandatory disclosures and produce informative research reports (Brown et al. 2015).

¹¹ We focus on equity analysts rather than bond analysts because equity analysts play a more important role in firms' capital structure. Equity analysts cover more publicly traded firms than bond analysts do, and their reports are readily available from data sets such as the Institutional Brokers' Estimate System (I/B/E/S) data. In the 2006 I/B/E/S data, 4,838 unique firms are covered by equity analysts, whereas De Franco et al. (2009) obtained bond analyst data for only 633 firms. Johnston et al. (2009) showed that debt analysts do not provide much incremental information for firms with low- or even medium-risk debt, whereas equity analysts cover firms with even modest bond values. When an equity analyst report is issued, the bond market exhibits an abnormal trading volume (De Franco et al. 2009). Bond analysts work closely with equity analysts (Lee 2002, Bond Market Association 2004, Ronan 2006). In our analysis, if equity analysts have no information role on bank debt and the bond market, it is bias against our findings.

¹² For example, Barth and Hutton (2004) demonstrate that greater analyst coverage helps stock prices to more rapidly incorporate

information on accounting accruals and cash flows, indicating that analysts improve market efficiency in evaluating financial statements.

¹³ Financial analysts produce information to capital markets, which helps investors value firms more efficiently. See surveys by Healy and Palepu (2001) and Beyer et al. (2010) for detailed reviews of the research on the role of financial analysts in capital markets.

¹⁴ To conserve space, we incorporated additional analyses such as robustness checks into the internet appendix.

¹⁵ Early research used debt structure data from Compustat, which does not provide detailed and accurate classification of debt. S&P Capital IQ classifies debt into commercial paper, revolving credit, term loans, senior bonds and notes, subordinated bonds and notes, capital leases, and other debt, allowing researchers to provide additional evidence of debt structure. Colla et al. (2013) point out that S&P Capital IQ's coverage has become more comprehensive since 2002.

¹⁶ Other debts and capital leases are excluded, as it is unclear whether these belong to private or public debt.

¹⁷ Our results do not change if we match the treated and control firms according to their debt structure in the year before the event.

¹⁸ The number of matched control firms ranges from 1 to 23. To avoid bias from an imbalance in the number of matched firms, we require our treated firms to have at most five matched firms. Our results do not change if we use the full sample, that is, with no restriction on the number of matched control firms.

¹⁹ We are grateful to an anonymous referee for suggesting we conduct this analysis.

²⁰ To save space, we report only the interaction terms in the table.

²¹ We also conduct robustness checks for the low-analyst-coverage subsample (below the median). As reported in Table IA4 of the internet appendix, the pattern of the findings on asset structure, issuance, and debt components in the full sample are all replicated in the subsample with low initial analyst coverage.

²² Given the sample selection criteria, the most risky firms in Bolton and Freixas's (2000) model are unlikely to appear in our sample. For example, our sample firms are larger than the average firm in Center for Research in Security Prices–Compustat universe.

²³ Colla et al. (2013, p. 2117) find that “the degree of debt specialization varies widely across different subsamples—large rated firms tend to diversify across multiple debt types, while small unrated firms specialize in fewer types.” Our sample contains mainly large firms, and it is worthwhile to provide more evidence on the choice among different types of debt instruments.

²⁴ Colla et al. (2013) document that about half of their sample firms use either drawn credit lines or term loans for financing. Acharya et al. (2014) propose a theory of credit lines provided by banks to firms as a form of monitored liquidity insurance, and their empirical evidence confirms the theory. Sufi (2009) documents that the total line of credit represents 16% of book assets, suggesting that lines of credit are widely used by public firms, and they represent large amounts of used debt and unused debt availability. Ivashina and Sharfstein (2010) find that during the subprime crisis, firms used more credit lines to ensure that they had access to funds at a time when there was widespread concern about the solvency and liquidity of financing options.

²⁵ We are grateful to an anonymous reviewer for suggesting we investigate trade credit.

²⁶ We find a similar pattern for the bond returns if we exclude data from 2008 to 2010.

²⁷ According to Chava and Roberts (2008), the DealScan database contains 50–75% of the value of all commercial loans in the United States during the early 1990s, and coverage increases after 1995 to include an even greater proportion of commercial loans.

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